

Serena Pei

Computer scientist building AI solutions with research depth and practical engineering experience
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EDUCATION

Massachusetts Institute of Technology (MIT)

2022 - 2026

GPA: 4.9/5.0

- Candidate for a Bachelor of Science degree in Computer Science and Engineering
- Gordon-MIT Engineering Leadership Program (GEL) student, Eta Kappa Nu (HKN) honors, Chinese minor
- Coursework: Computation Structures, Linear Algebra, Machine Learning, Elements of Software Construction, Design and Analysis of Algorithms, Operating System Engineering, Computer Systems Engineering, Computer Vision, Natural Language Processing, Distributed Computer Systems Engineering, Probability, Statistical Data Analysis, Deep Learning, ML for Healthcare

SKILLS/TOOLS

Skills: Python, Java, Typescript, Javascript, Node.js, PostgreSQL, MongoDB, Unix, C, Assembly, Selenium, R, React, Flask, HTML, PyTorch, Scikit-learn, Data Structures, Algorithms, Data/ML/Security Infrastructure, System Design, Computer Vision, LLMs, REST API

Tools: Git/Github, AWS, Google Cloud, VS Code, Jupyter Notebook, IntelliJ, Google Colaboratory, LaTeX, Docker, MS Office

EXPERIENCE

MIT Media Lab (Fluid Interfaces Lab)

Cambridge, MA

Research Assistant under Dr. Nataliya Kosmyna (nkosmyna@media.mit.edu)

Feb 2024 - Present

- Co-author of paper accepted to 2024 IEEE 20th International Conference on Body Sensor Networks (BSN), study evaluated machine learning model prediction of eye movement from EOG and EEG signal data (700+ lines of Python)
- First author of study using Electromyography (EMG) signals for minimally verbal ASD individuals (1000+ lines of Python)
- Co-author of paper accepted for publication in ACM IMWUT, designed and conducted experiments on participants, examined emotion regulation through heartbeat recognition and valence/arousal disturbance with statistical tests
- Led multimodal analysis study using EOG/EEG/EMG signal data for ML model prediction, onboarded and mentored new students, assigned tasks and organized meetings to keep track of progress, wrote 2000+ lines of Python code for analysis

Amazon (People eXperience Technology)

Seattle, WA

Software Development Engineer Intern (Machine Learning)

May 2025 - Aug 2025

- Designed and implemented from scratch a feedback mechanism with a data processing pipeline and full stack web application to incorporate human-in-the-loop and LLM capabilities for improving a RAG-based chatbot system
- Documented research results and code base, presented final project and demo to stakeholders and science team members
- Used AWS CDK to orchestrate a state machine of lambda functions and a secure internally authenticated full stack web application (pushed 60000+ lines of Python, Javascript, and HTML/CSS to the Amazon internal code system)

MIT Computer Science and Artificial Intelligence Laboratory (FutureTech Lab)

Cambridge, MA

Research Assistant under Dr. Wensu Li (wensu@mit.edu)

Mar 2024 - May 2025

- Analyzed the economics of labor automation using AI and fit functional forms of computer vision model scaling laws
- Wrote 1000+ lines of Python code to run scaling law experiments, visualize data, and analyze survey results
- Used prompt engineering to produce robust estimations on classification complexities for automating occupations

A*STAR (Agency for Science, Technology, and Research) Bioinformatics Institute

Singapore

Intern under Dr. Lau Mai Chan (lau_mai_chan@bii.a-star.edu.sg)

May 2024 - Aug 2024

- Implemented unsupervised model for feature extraction and tissue clustering for histomorphology in H&E stained tissue images from liver cancer patients (5000+ lines of Python code for model training and evaluation, and data visualization)
- Conducted literature reviews for SOTA deep learning models, clustering methods, and self-supervised encoders
- Optimized code for opening high resolution whole slide images, overlaid predicted cluster masks to analyze accuracy

MIT Computer Science and Artificial Intelligence Laboratory (Kellis Lab)

Cambridge, MA

Undergraduate Research Assistant under Manolis Kellis (manoli@mit.edu)

Dec 2023 - Apr 2024

- Web scraped data from various websites using Selenium and embedded data to feature vectors using Python
- Mapped over 3000+ classes and majors from courseroad, 900+ courses from EdX, and 400+ MIT job descriptions on an AI navigator space (seascapes.ai) using PostgreSQL

TEACHING

MIT Introduction to Machine Learning (6.3900)

Cambridge, MA

Learning Assistant

Jan 2025-Dec 2025

- Staffed office hours for homework, lab checkoffs, and exam review; contributed to staff meetings to improve assignments, overall class structure, and discuss content (7+ hrs/wk)

Cover Letter

Dear SERC Selection Committee,

As a senior studying Computer Science and Engineering with extensive research experience in AI and human-computer interaction, I have consistently sought to bridge technical innovation with profound human-centered questions.

I'm writing to express my enthusiastic interest in the SERC Scholars Program, particularly the project "The Computational Roots of Human Suffering." This project deeply resonates with my academic and personal journey, where I've explored how technology can better understand and serve human needs, from building studying biosignal-driven interfaces at the MIT Media Lab to implementing cancer diagnosis driven ML pipelines in Singapore.

I'm also drawn to the "Synergy" project's focus on how human and AI collaboration can enhance collective intelligence rather than replace human ingenuity. Similarly, my previous project experience in the FutureTech Lab aimed to explore the balance between automation and augmentation of occupations by computer vision models. I'm excited to continue this exploration by using state-of-the-art techniques in AI.

The "Aligning AI with Human What?" project's focus on the foundational question of what human values truly entail interests me because I hope to explore the conceptual underpinnings of what lies behind the technologies we use everyday. Going forward, I'm eager to help identify gaps between theoretical ideals and practical implementations from an ethical perspective.

Through this program, I hope to gain a richer, interdisciplinary understanding of how computational frameworks can address deeply human problems like suffering, emotional regulation, and cognitive biases. I am eager to learn from contemplative practices and theoretical discussions, and to integrate these insights into my technical work, ultimately developing tools that are both powerful and ethically grounded.

I'm excited to contribute my skills in AI/ML research, data analysis, and experimental design to investigate computational models of human suffering. I'm particularly interested in developing projects that combine computational modeling with therapeutic applications in order to create practical tools to encourage mental wellbeing.

I hope to engage deeply with the weekly seminars, experiential practice, and collaborative discussions with peers and mentors across disciplines. Thank you for your time for considering my application.

Sincerely,

Serena Pei